

Meanings (concise)

- **betavis**: Fraction of VIS shortwave treated as surface-absorbed (augments NIR-based beta for lakes).
- **cnfac**: Crank–Nicolson time-centering factor (0–1) for diffusion solves.
- **cpliq**: Specific heat capacity of liquid water [$\text{J kg}^{-1} \text{K}^{-1}$].
- **crop**: Per-PFT flag array; 1 = crop PFT, 0 = non-crop PFT.
- **denh2o**: Density of liquid water [kg m^{-3}].
- **dtime_mod**: Model time step length [s] (default 3600).
- **grav**: Gravitational acceleration [m s^{-2}].
- **hfus**: Latent heat of fusion of ice [J kg^{-1}].
- **iscft**: Not present; generally used to index/flag crop functional types (CFTs).
- **lake_no_ed**: If true, disable eddy-diffusion mixing contribution in lake thermal diffusion.
- **lakepuddling**: If true, suppress convection when sufficient ice is present; enables “puddling” behavior.
- **nlevgrnd**: Number of ground (soil+bedrock) temperature layers.
- **nlevlak**: Number of lake temperature layers.
- **nlevsno**: Maximum number of snow layers.
- **nlevsoi**: Number of hydrologically active soil layers.
- **percrop**: Per-PFT flag array; 1 = perennial crop PFT, 0 = not perennial crop.
- **thk_bedrock**: Thermal conductivity assigned to bedrock [$\text{W m}^{-1} \text{K}^{-1}$].
- **tkwat**: Thermal conductivity of water [$\text{W m}^{-1} \text{K}^{-1}$].
- **use_lch4**: Enable CH₄ coupling/diagnostics (affects lake resistance and CH₄ conductance).
- **vk**: von Kármán constant (dimensionless).

Here are practical ranges and typical/default values you can assume for unit tests:

- **betavis**: [0, 1] (dimensionless). Default 0.0.
- **cnfac**: [0, 1] (dimensionless Crank–Nicolson weighting). Default 0.5.
- **cpliq**: ~4180–4190 $\text{J kg}^{-1} \text{K}^{-1}$ (specific heat of liquid water). Model uses a constant from SHR constants (~4182).
- **crop**: 0 or 1 (per-PFT flag; 1 = crop).
- **denh2o**: ~1000 kg m^{-3} (liquid water density). Constant in model (~1000).
- **dtime_mod**: > 0 s; typical 1800–3600 s. Default 3600 s.
- **grav**: ≈ 9.806 m s^{-2} (Earth gravity).
- **hfus**: ≈ $3.34 \times 10^5 \text{ J kg}^{-1}$ (latent heat of fusion of ice).
- **iscft**: Not defined in this tree; if used elsewhere it’s typically an integer index 1..num_cfts or a logical flag.
- **lake_no_ed**: logical (.true./.false.). Default .false.
- **lakepuddling**: logical (.true./.false.). Default .false.
- **nlevgrnd**: positive integer. Typical defaults 15 or 15 + nlev_equalspace (with nlev_equalspace=15 → 30).
- **nlevlak**: positive integer. Default 10; 25 if use_extralakelayers = .true.

- `nlevsno`: positive integer. Default 5; 16 if `use_extrasnowlayers = .true.`
- `nlevsoi`: positive integer. Either `nlevsoifl` (from input) or `8 + nlev_equalspace` ($\rightarrow$ 23 with default `nlev_equalspace=15`).
- `percrop`: 0 or 1 (per-PFT flag; 1 = perennial crop).
- `thk_bedrock`: $\approx 3.0 \text{ W m}^{-1} \text{ K}^{-1}$ (thermal conductivity of bedrock). Constant.
- `tkwat`: $\approx 0.57 \text{ W m}^{-1} \text{ K}^{-1}$ (thermal conductivity of water). Constant.
- `use_lch4`: logical (.true./.false.). Default .false.
- `vk`: ≈ 0.40 (von Kármán constant, dimensionless). Constant.

Here are compact boundary-value sets you can use to exercise key branches in LakeTemperature. Use one set per test; combine where helpful. Values not shown can remain at model defaults.

- Defaults (sanity baseline)
 - `betavis=0.0`; `cnfac=0.5`; `lake_no_ed=.false.`; `lakepuddling=.false.`; `use_lch4=.false.`
 - `nlevlak=10`; `nlevgrnd=15`; `nlevsoi=nlevsoifl`; `nlevsno=5`
 - `dtmod=3600.0`
 - `cpliq $\approx$ 4182`; `denh2o $\approx$ 1000`; `tkwat=0.57`; `thk_bedrock=3.0`; `grav $\approx$ 9.806`; `vk $\approx$ 0.40`
 - `crop=0`; `percrop=0`
- Time discretization extremes (numerics)
 - Explicit: `cnfac=0.0`
 - Fully implicit: `cnfac=1.0`
- Shortwave partition extremes (radiation into lake)
 - All VIS treated as VIS (max penetration via beta's VIS term): `betavis=0.0`
 - All VIS treated as NIR at surface (min penetration): `betavis=1.0`
- Eddy diffusion on/off (mixing pathway)
 - Eddy mixing enabled: `lake_no_ed=.false.`
 - Eddy mixing disabled: `lake_no_ed=.true.`
- Puddling logic toggle (convection suppression under ice)
 - Off: `lakepuddling=.false.`
 - On: `lakepuddling=.true.` (Pair with `ice  $\geq$  pudz` in state to trigger path; constants-only test simply toggles the switch.)
- Methane coupling (resistance/cond calculations)
 - Off: `use_lch4=.false.`
 - On: `use_lch4=.true.`
- Vertical resolution (affects diffusion, interfaces, convection)
 - Minimal safe lake layering: `nlevlak=2`
 - Deep-lake option: `nlevlak=25`
 - Ground layers: `nlevgrnd=15` and `30`
 - Snow layers cap: `nlevsno=5` and `16`
 - Soil active layers: `nlevsoi=min(nlevsoifl, 8+15)` and `8+15`
- Time step extremes (stability/energy accounting)
 - Small: `dtmod=60.0`
 - Large (still reasonable): `dtmod=7200.0`
- Physical constants (robustness; keep near-physical)
 - `cpliq`: nominal (4182), and $\pm 5\%$ perturbation
 - `denh2o`: nominal (1000), and $\pm 2\%$ perturbation
 - `tkwat`: nominal (0.57), and $\pm 10\%$ perturbation
 - `thk_bedrock`: nominal (3.0), and $\pm 20\%$ perturbation

- vkc: 0.35 and 0.45 (around 0.40)
- grav: 9.70 and 9.86 (around 9.806)
- hfus: keep nominal; $\pm 1\%$ if you must test sensitivity
- Crop flags (not used by LakeTemperature, but for completeness)
 - crop $\in \{0,1\}$; percrop $\in \{0,1\}$
 - iscft: not present here; if used elsewhere, test $\{1, \text{num_cfts}\}$

Example test tuples (only list changed values):

- Mixing off: {lake_no_ed=.true.}
- Puddling on: {lakepuddling=.true.}
- Radiative max-surface: {betavis=1.0}
- CN explicit: {cnfac=0.0}
- Minimal layering: {nlevlak=2, nlevgrnd=15, nlevsno=5}
- Deep layering: {nlevlak=25, nlevgrnd=30, nlevsno=16}
- Short time step: {dtime_mod=60.0}
- CH4 on: {use_lch4=.true.}

If you want, you can assemble a small parameterized test matrix combining these into ~10 high-yield cases that collectively cover all branches.

Here's a compact 10-case matrix that hits the main branches/sensitivities in LakeTemperature. Use one row per test; unspecified values can stay at defaults.

- Baseline sanity
 - {betavis=0.0, cnfac=0.5, lake_no_ed=.false., lakepuddling=.false., use_lch4=.false., nlevlak=10, nlevgrnd=15, nlevsoi=nlevsoifl, nlevsno=5, dtime_mod=3600}
- Radiative surface-absorption extreme
 - {betavis=1.0}
- Explicit time-centering (stability edge)
 - {cnfac=0.0, dtime_mod=3600}
- Fully implicit time-centering
 - {cnfac=1.0, dtime_mod=3600}
- Eddy mixing disabled (molecular + optional Fang-Stefan only)
 - {lake_no_ed=.true.}
- Puddling logic enabled (convection suppression under ice)
 - {lakepuddling=.true.}
 - Pair with state where total lake ice thickness $\geq \text{pudz}$ to exercise suppression branch.
- Methane coupling on (resistance and conductance path)
 - {use_lch4=.true.}
 - Pair with unfrozen top layer ($\text{lake_icefrac}(1) \leq 0.1$) to avoid CH4 cutoff.
- Minimal lake layering (interface math edge)
 - {nlevlak=2, nlevgrnd=15, nlevsoi=nlevsoifl, nlevsno=5, dtime_mod=3600}
- Deep layering + long timestep (diffusion, mixing scaling, performance)
 - {nlevlak=25, nlevgrnd=30, nlevsoi=8+15, nlevsno=16, dtime_mod=7200}
- Physical constants perturbation (robustness/sensitivity)
 - {cpliq=1.05×default, denh2o=1.02×default, tkwat=0.9×default, thk_bedrock=1.2×default, vkc=0.35, grav=9.70, hfus=1.01×default}

Optional add-ons to combine with any case as needed:

- Frozen/no-snow vs snow-on cases to drive different solar source splits: ensure one test with $snl=0$, $t_lake(1)>tfrz$; one with $snl<0$; one with $t_lake(1)<tfrz$.
- Crop flags (should be inert here): $\{crop=1, percrop=1\}$ to confirm no coupling to lake thermodynamics.